

Case Study by Jex De Los Santos
on September 2, 2026

Tools Used

- Google Sheets
- Python
- PostgreSQL
- DBT
- Github
- PowerBI
- MS Powerpoint

Business problem

For a bike-share company, they have customers that are casual bikers who are one-time users of their bike, and there are members who are part of the annual membership. The problem for this company is casual riders are not interested in getting annual memberships.

The task is to find insights on how casual riders are different from annual members and use them to create a plan to convert casual riders into annual subscribing members.

About the data

The data is originally from Divvy accessed from their Amazon AWS:
<https://divvy-tripdata.s3.amazonaws.com/index.html>

The data does not contain Personally Identifiable Information and it is officially owned by the City of Chicago. Given that the data is of type first-party data, the data credibility is strong. Additionally, the data integrity is fixed by cleaning later on in this document.

The data selected to analyze is from August 2025 up to June 2026, each month contained in a separate zip file. Each zip file contains the data in .csv format.

Upon examining the data in a spreadsheet, it contains empty values for the *station* columns.

Data cleaning

Using Python, I:

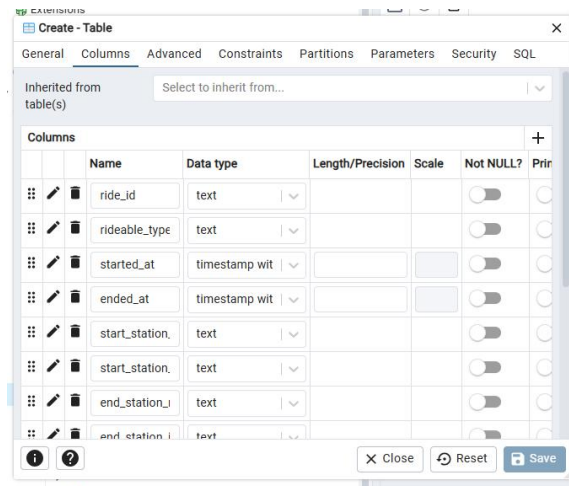
1. Removed the null values
2. Removed the duplicates
3. And combined the 12 .csv files into 1 .csv file

Here is the link to Kaggle, the Python notebook I used to manipulate the data:

<https://www.kaggle.com/code/x00000/cleaning-bike-data>

The data is now combined into 1 .csv file. The file size is 781MB.

The csv data is then ready to be uploaded into the following PostgreSQL table.



After importing the data into PostgreSQL, the data is ready to be queried.

The data is then queried from the PostgreSQL database using DBT as connector. These DBT SQL queries are models where each query creates a View. The queries I used that created the PostgreSQL views is available here in Github:

https://github.com/jexdls/data-analysis/tree/main/bikers_analysis/models/example

Upon examining the data in PostgreSQL views, I found out that the column *started_at* has values that are earlier than *ended_at*. This is not possible since you can only end a trip at a date time later than the start date time. This created a problem where *trip_durations* have negative values. Since, *trip_duration* pertains to how long the ride lasted, it should not be possible to have a negative value so I investigated further.

started_at	ended_at
timestamp with time zone	timestamp with time zone
2025-11-02 01:59:56.801+08	2025-11-02 01:05:09.113+08
2025-11-02 01:57:24.553+08	2025-11-02 01:02:44.165+08
2025-11-02 01:59:07.835+08	2025-11-02 01:05:19.214+08
2025-11-02 01:55:38.177+08	2025-11-02 01:04:35.315+08

Another problem was found where thousands of records has *trip_durations* of less than 1 second. Which should not also be possible.

trip_duration interval	is b
00:00:00.686	t
00:00:00.697	t
00:00:00.703	t
00:00:00.711	t
00:00:00.712	t
00:00:00.717	t
00:00:00.719	t
00:00:00.72	t
00:00:00.72	t
00:00:00.723	t
00:00:00.729	t
00:00:00.732	t

To improve data integrity, I thought I have to solve these problems first and delete rows. If I'm going to delete rows whose *trip_duration* is not valid, what *trip_duration* value should be considered valid? Negative *trip_durations* are surely not valid. Even millisecond trips are not valid, but where should I stop? Are 1-second or 5-second trips valid or should be deleted? While looking for the limit, I noticed that some *start_stations* and *end_stations* are the same for trips that lasted less than a minute. However, the *start_station* being the same as *end_station* is not enough of a basis to say that the record is not valid because there are records that lasted less than a second whose *start_stations* are not *end_stations*.

	ride_id text	start_station_id text	end_station_id text	trip_duration interval
1	66E6BFC148F324BF	CHI00234	CHI00834	00:00:00.466
2	430F4E31C86ACACA	CHI02069	CHI02058	00:00:00.717
3	4AA621A32C615BD9	CHI00234	CHI00834	00:00:00.888
4	51B79AD29CBD7C57	CHI00234	CHI00834	00:00:00.899
5	FEE6AAAABC451EA01	CHI01015	CHI01592	00:00:00.967
6	EDEAE7D22F819FA8	CHI00266	CHI01540	00:00:01.167
7	154396CB31A28BBE	CHI02183	CHI01504	00:00:01.184
8	3CD70F125669ED5D	CHI02055	CHI02063	00:00:01.237
9	FC68C6AA2B6AD650	CHI02292	CHI02291	00:00:01.246
10	2F7CAE8D7CAF4CCF	CHI02055	CHI02063	00:00:01.258
11	448951D36113A8D9	CHI00234	CHI00834	00:00:01.269
12	C1BCC13FB994EC85	CHI02055	CHI02063	00:00:01.29
13	0988AFBA9F8A8DDE	CHI02122	CHI02065	00:00:01.317

Upon investigating further, the following query revealed the reason. I first looked at trips that started and ended at different stations to rule out trips that started and ended at the same stations.

```

1  SELECT
2      ride_id,
3      start_station_id,
4      end_station_id,
5      trip_duration,
6      member_casual,
7      rideable_type
8  FROM
9      schema.new_columns
10 WHERE
11     start_station_id <> end_station_id
12 ORDER BY
13     trip_duration ASC
14 LIMIT 5000

```

It was revealed that these extremely short trips are of type *electric_bike* which could mean that *electric_bikes* have a bug that registers these extremely short trips into the database.

	ride_id text	start_station_id text	end_station_id text	trip_duration interval	member_casual text	rideable_type text
1	66E6BFC148F324BF	CHI00234	CHI00834	00:00:00.466	casual	electric_bike
2	430F4E31C86ACACA	CHI02069	CHI02058	00:00:00.717	member	electric_bike
3	4AA621A32C615BD9	CHI00234	CHI00834	00:00:00.888	member	electric_bike
4	51B79AD29CBD7C57	CHI00234	CHI00834	00:00:00.899	member	electric_bike
5	FEE6AAABC451EA01	CHI01015	CHI01592	00:00:00.967	casual	electric_bike
6	EDEAE7D22F819FA8	CHI00266	CHI01540	00:00:01.167	casual	electric_bike
7	154396CB31A28BBE	CHI02183	CHI01504	00:00:01.184	member	electric_bike
8	3CD70F125669ED5D	CHI02055	CHI02063	00:00:01.237	casual	electric_bike
9	FC68C6AA2B6AD650	CHI02292	CHI02291	00:00:01.246	casual	electric_bike
10	2F7CAE8D7CAF4CCF	CHI02055	CHI02063	00:00:01.258	member	electric_bike
11	448951D36113A8D9	CHI00234	CHI00834	00:00:01.269	member	electric_bike
12	C1BCC13FB994EC85	CHI02055	CHI02063	00:00:01.29	member	electric_bike
13	0988AFBA9F8A8DDE	CHI02122	CHI02065	00:00:01.317	member	electric_bike

For the other type of bike, when the *trip_duration* for *classic_bikes* are sorted from least, the lowest *trip_duration* is 01:00:087, which is at 1 minute mark. This means that *classic_bike* trips are more reliable than *electric_bikes*.

	ride_id text	start_station_id text	end_station_id text	trip_duration interval	member_casual text	rideable_type text
1	0C649C18521F6E17	CHI00440	CHI00501	00:01:00.087	member	classic_bike
2	AA73952EDCB96B90	CHI00453	CHI00449	00:01:00.172	member	classic_bike
3	6C2D792F7579A1F5	CHI00750	CHI00332	00:01:00.185	member	classic_bike
4	E87E41AA79695C15	CHI00321	CHI00406	00:01:00.186	member	classic_bike
5	D6C50CD9ED1992D1	CHI01928	CHI01745	00:01:00.277	member	classic_bike
6	51847E0D22F36CAD	CHI00438	CHI00405	00:01:00.299	member	classic_bike
7	372E14BCF7DE387E	CHI00288	CHI00555	00:01:00.333	member	classic_bike
8	50058E8987F18A66	CHI00418	CHI00470	00:01:00.336	member	classic_bike
9	FCC6CE119DCA42CF	CHI00233	CHI02196	00:01:00.336	member	classic_bike
10	4A5DDC43278082F7	CHI00470	CHI01751	00:01:00.35	member	classic_bike
11	D8581F11B46E699E	CHI01161	CHI00240	00:01:00.355	member	classic_bike
12	B649C5B732A0A3CF	CHI00521	CHI02193	00:01:00.402	member	classic_bike
13	7A0C60A6C1BC1039	CHI02132	CHI02134	00:01:00.406	member	classic_bike

To further investigate, the following query was made to look at the minimum *trip_duration* for each bike type.

```

1  SELECT
2      MIN(trip_duration),
3      rideable_type
4  FROM
5      schema.new_columns
6  WHERE
7      trip_duration >= '00:00:00'
8      AND start_station_id <> end_station_id
9  GROUP BY
10     rideable_type
11  LIMIT 5000

```

Data Output

Messages

Notifications

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	min interval	rideable_type text
1	00:01:00.087	classic_bike
2	00:00:00.466	electric_bike

It was revealed that the minimum *trip_duration* for *electric_bike* are 0 second trips, which may be concluded that valid trips should instead start from the minimum *trip_duration* for *classic_bikes* which is 00:01:00.87 which is more reliable. To fix the

data integrity, I deleted all the records that has a *trip_duration* of less than 00:01:00.87 because they aren't reliable.

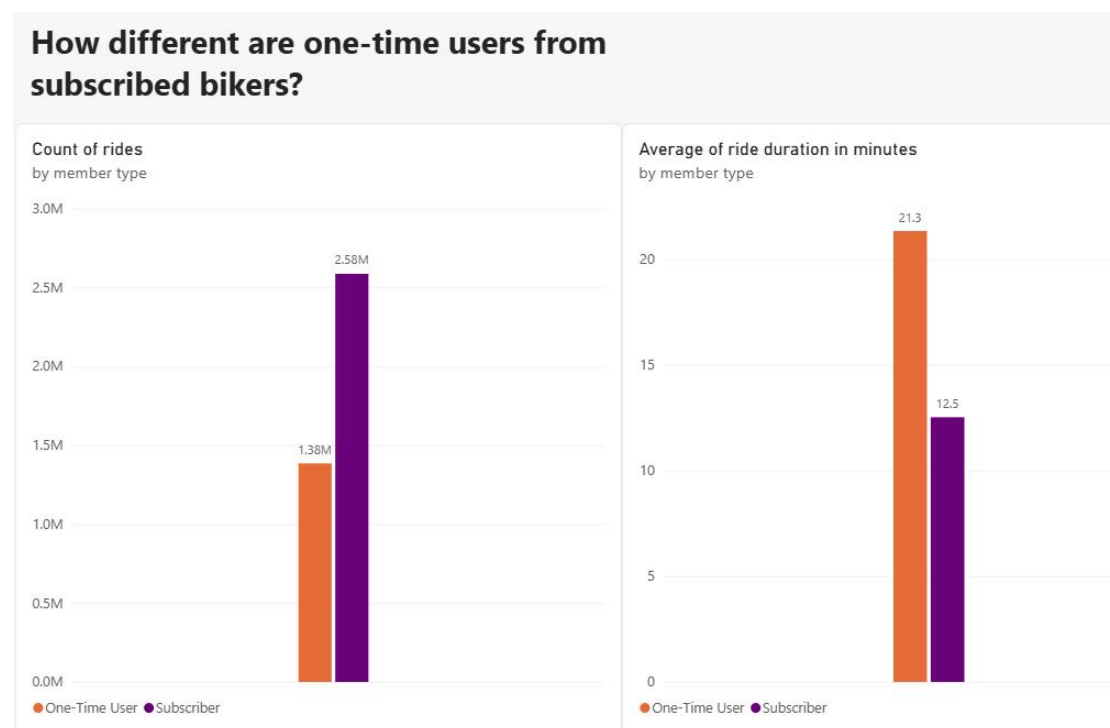
Insights and Findings

After all the cleaning, the data is ready for visualizing. Through Power BI, the visualizations were created.

To see the summarized findings ready for a presentation please download the file here:

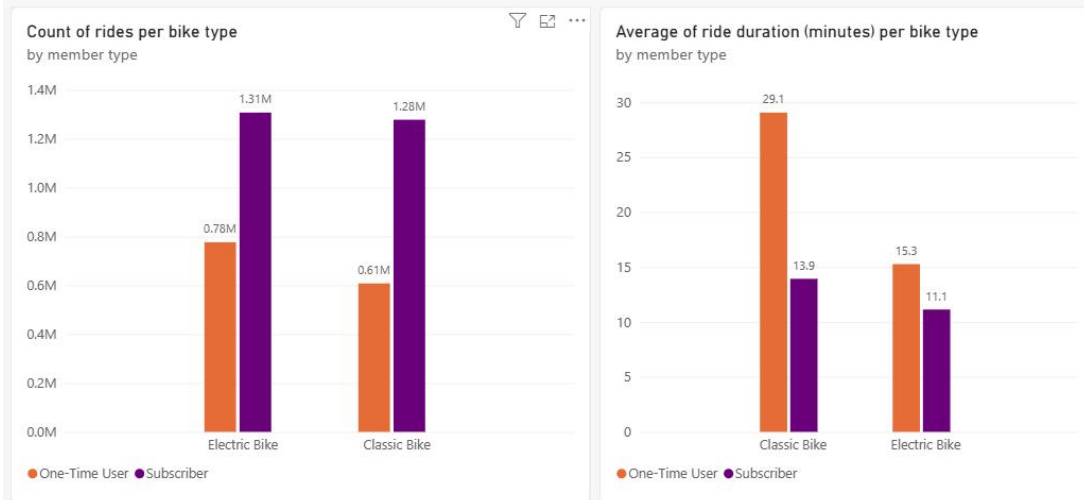
https://github.com/jexdls/data-analysis/blob/main/Bikers_analysis.pptx

The following examines the findings in greater detail.



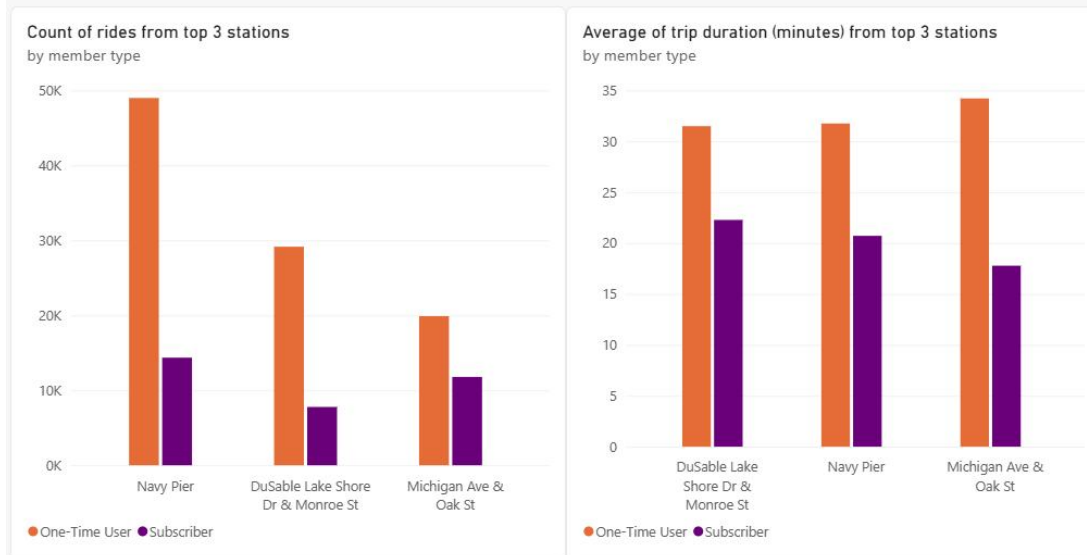
How different are one-time users from subscribed bikers? According to the data, 2.58M of bikers are from subscribed bikers. The casual bikers are almost half of the subscription bikers with 1.38M bikers. Although there are more subscription bikers, casual bikers tend to have longer rides than subscription bikers. They take an average of 21.3 minutes of riding while the subscribed bikers take almost half of these bikers having an average of 12.5 minutes ride.

How different are one-time users and subscribed bikers when using different types of bike?



How different are one-time users from subscribed bikers when using different types of bike? There's no significant difference for this. Both casual bikers and subscription bikers have a longer average ride on classic bikes. Additionally, there's more number of rides for electric bikes than classic bikes.

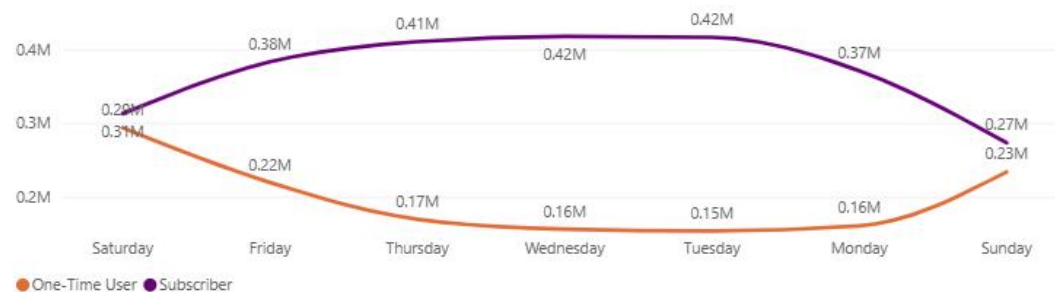
How different are the one-time users and subscribed bikers of the top 3 most used station?



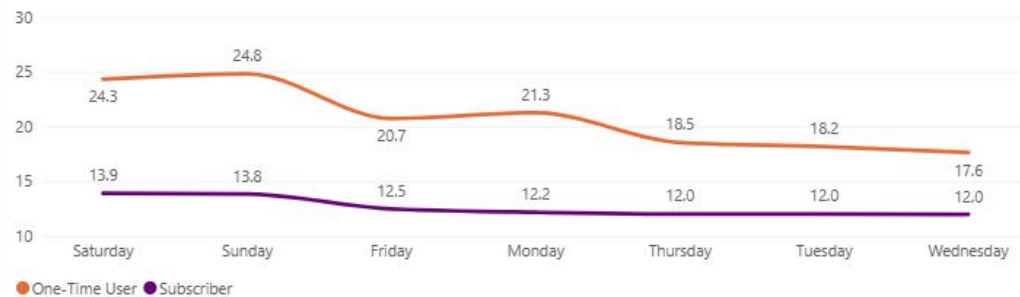
How different are the one-time users and subscribed bikers who start from the top 3 most used stations? One-time users have more number of rides as well as have longer rides than subscribed bikers in the top 3 stations most rides start at which are Navy Pier, Michigan Ave & Oak St., and DuSable Lake Shore Dr & Monroe St.

How different are one-time users from subscribed bikers on their weekly rides?

Count of rides per day of week
by member type



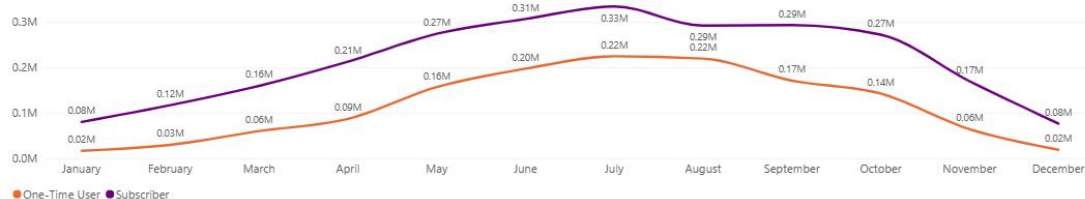
Average of ride duration (minutes) per day of week
by member type



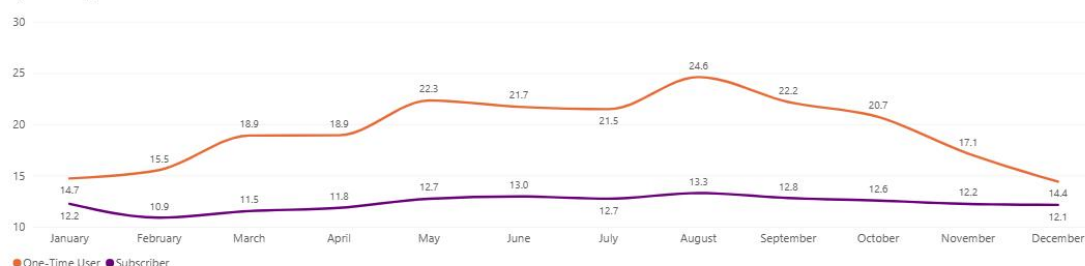
How different are one-time users from subscribed bikers on their weekly rides? During weekdays, there is an increase of rides from subscribed bikers, and a decrease of rides from one-time users. However, there is an increase of one-time rides during the weekends. Additionally, both one-time users and subscribed bikers have longer rides during the weekends.

How different are one-time users and subscribed bikers on their monthly rides?

Count of rides per month
by member type

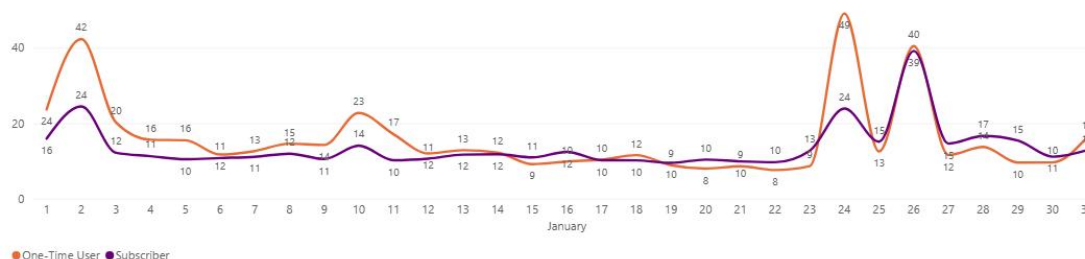


Average of ride duration (minutes) per month
by member type



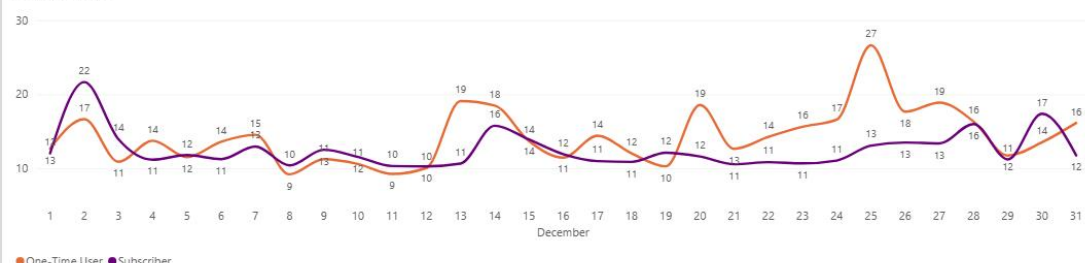
How different are one-time users and subscribed bikers on their monthly rides? There is an increase of rides from May to October. Additionally, one-time users have longer rides from May to October while subscribed bikers have a sustained duration of rides throughout the year.

Average of ride duration (minutes) per month
by member type



For a detailed view on January, there is a sudden increase of rides having longer duration on January 2. This is likely caused by the celebration of New year.

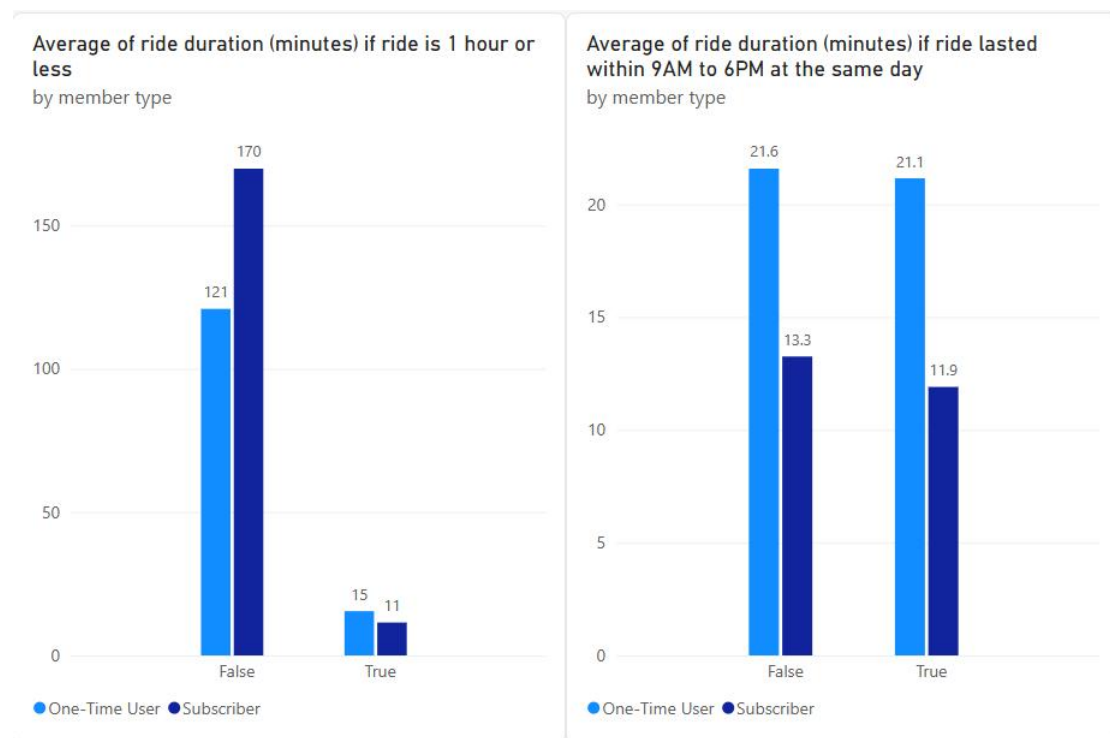
Average of ride duration (minutes) per month
by member type



For a detailed view on December, there is also a sudden increase of one-time rides having longer duration on December 25. This is likely caused by the celebration of Christmas. On the other hand, subscribers have a sustained duration of rides throughout the day.



For a detailed view on February, there is also a sudden increase of one-time rides having longer duration on February 14, 15, and 16, which may be caused by the celebration of couples during Valentines day. On the other hand, subscribers have a sustained duration of rides throughout these days.



Furthermore, subscribers have longer rides for rides that take more than an hour while one-time users have shorter rides for rides that take more than an hour. This can speak of subscribers making the most out of their subscription while one-time users appear to save for long rides given that one-time users tend to want longer rides.

Next, one-time users have longer rides for rides that take 1 hour or less while subscribers have shorter rides for rides that take 1 hour or less.

Lastly, one-time users tend to have longer ride duration whether rides are within 9AM to 6PM or not. There is no significant difference for these rides.

Conclusion

It can be inferred that the one-time users primarily use the bikes for leisure and want longer rides while annual members show a more routinized use of bikes and save for longer rides. This is useful for commuting, or even simply going to work.

Recommendation

Since one-time users may want longer rides for leisure it is recommended to create member-only offers that save them money for long rides during special days like Christmas.

Promote these offers in the top 3 stations Navy Pier, Michigan Ave & Oak St., and DuSable Lake Shore Dr & Monroe St. to maximize reaching one-time users.

Lastly, create enticing offers that are available for members during the weekends.